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Call for the mainstreaming of eutrophication in global water assessments and policies

Olli Varis & Dandan Zhao

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This comment prompts scholars to include eutrophication more often to water quality assessments as it is globally one of the major water-related environmental problems but remains underrepresented in contemporary assessments, water policies and international targets such as the United Nations Sustainable Development Goals (SDGs). What we scholars can do is to mainstream eutrophication in our global water assessments, and that way ram it to a higher position in political agendas. The recent advances in global nutrient models and datasets, as well as Earth observation approaches, allow novel horizons in this regard. Now is the time to scrutinize these novel opportunities. Our take to this quest here is to develop a model protocol that produces gridded eutrophication maps from total nitrogen and total phosphorus data. We also developed analogous methods for biochemical oxygen demand and total dissolved solids.

Elevated attention to eutrophication is needed in research and policy

Eutrophication is one of the major water quality problems of surface waters globally^{1–3}, causing massive deterioration of aquatic ecosystems, habitats, and biodiversity as well as remarkable environmental nuisance to human water uses. Eutrophication stands for enrichment of plant nutrients, most importantly nitrogen and phosphorus, resulting in increased biomass production^{4,5}. Eutrophication is a natural phenomenon, but human activities, such as discharge of waste and wastewater, and land use change result in increased erosion and leaching of nutrients. This greatly accelerates the input of nutrients to waters across the world, causing nuisance in terms of excessive biomass of algae and vegetation⁶. This, in turn, is prone to lead to an enhanced degradation of biomass, often leading to hypoxia. Warming of climate enhances the severity of these problems.

Whereas eutrophication has been identified and recognized as a major problem for over a century^{7,8}, and being recently highlighted within the Earth system boundaries context⁹, it keeps worsening⁹. This despite of

growing attempts to control non-point source pollution, together with massive and escalating investments in treatment of waste waters across the world, exceeding 300 billion USD annually¹⁰. Yet still 28% of the world's population had no access to wastewater treatment in 2020¹¹. Climate change further accelerates eutrophication¹².

Appreciating the progress of the SDG 6.2 on wastewater treatment and the SDG 6.3 on halving the proportion of untreated wastewater¹³, Jones et al.¹⁴ call for more ambitious development goals than those we see at present to curtail the further deterioration of water quality. To bear out this concern, eutrophication of freshwaters is gaining only occasional mention in UN Agenda 2030 reports, and is absent from the SDGs¹³. Alike, eutrophication has been virtually missing from global freshwater models and related assessments, although analyses and scenarios are now emerging^{15–17}, with initiatives to incorporate eutrophication into global data and model portals such as the ISIMIP¹⁸. At the same time, we see a progress in estimating the level of eutrophication from Earth observation data^{19–21}. Hopefully, these pioneering works will attract others within the modelling community to pay elevated attention to eutrophication since rigorous approaches are needed.

In general, the key factors that contribute to freshwater eutrophication are total nitrogen (TN) and total phosphorus (TP), being the most important drivers of the abundance of phytoplankton and the eutrophic condition of a water body²². With growing number of gridded datasets and models on concentrations of TN and TP in surface inland waters that have recently become available^{6,18,23–25}, making it possible to understand and quantify the global patterns and dynamics, their temporal evolution and driving factors of the fate of these key nutrients in aquatic systems on higher spatial resolution and scales. Thus, inspired by increasing water-related nutrient studies and datasets, this study makes this contribution by proposing a eutrophication model protocol that is implementable to gridded global and regional models employing TN and TP data. Such datasets allow a building ground for relatively straightforward ways to assess eutrophication.

We propose one based on the N and P co-limitation concept, and not solely on P limitation as has been common thus far¹⁷. It produces gridded eutrophication estimates using any spatial TN and TP datasets. We used the data of van Vliet et al.²⁵ on a $0.5^\circ \times 0.5^\circ$ resolution for the 2000–2010 period to investigate and illustrate how to create water quality classification-based eutrophication maps globally.

We employ the water quality classification scheme, an approach that has mainstreamed policy documents in eutrophication over decades^{26–28}. It is based on the Trophic State Index (TSI) concept²⁹, in which the waters are classified according to their primary production level. We derive this index from TN and TP concentrations as has been done in the above-mentioned documents, but today it can also be derived from Chlorophyll-a concentration observable visually, e.g., from remote sensing data²¹. TSI instead of concentrations makes sense because ecosystem response to increased P and N concentrations (as well as many other pollutants), and consequently

the adverse effect to water uses and public concerns, are typically not linear. Therefore, it is of prime importance to provide estimates on the harmfulness of eutrophication (expressed as TSI²⁹).

This protocol, for the first time, allows the production of quantified mappings of the harm due to freshwater eutrophication across the planet, under co-limitation of TN and TP. TSI has recently, though, been assessed from Landsat data, but has not been linked to nutrient concentrations²¹. As a side product of our eutrophication protocol, we propose analogous models for Biological Oxygen Demand (BOD) and salinity (total dissolved solids; TDS), documented in Supplementary Information. Such models are an important step in mainstreaming the consideration of eutrophication, BOD and TDS in large-scale water models and assessments.

The three-step protocol for eutrophication modelling

From TN and TP concentrations to eutrophication classification. First, the TN and TP concentrations are transferred into eutrophication classes (Fig. 1) using TSI²⁹. Typically, the classes used are ultra-oligotrophic (very low), oligotrophic (low), mesotrophic (intermediate), eutrophic (high), and hypertrophic (very high), implying an increasing level of primary production and thus eutrophication. The eutrophication literature of the past several decades abounds with various water quality

classifications^{26,27,30}. They vary across various agencies and geographic locations, and specific classifications exist for numerous fractions of nutrients and different types of water bodies, such as rivers, streams, deep lakes, shallow water bodies and more³¹.

Nevertheless, many water quality guidelines for policy and management purposes are reasonably consistent with one another. Based on the most important of them^{26–28,30}, we produced the values presented in Table 1 for the purposes of our study.

To develop equations for transferring nutrient concentrations into the scale of water quality classifications of Table 1, we used logarithmic regression analysis and further scaled the transformed maps into the unity scale so that the best water quality receives the index value 0 and the worst one 1. Instead of a categorical value from 1 to 5 of the conventional classifications, it gives a continuous number [0, 1], which is much more useful in, for instance, hazard and risk analysis as recommended in many contemporary United Nations protocols^{32,33}. Here, we need to assign minimum values for TN and TP ($TN_{min} = 0.1$ mg/L and $TP_{min} = 0.001$ mg/L). This is done because the used gridded datasets include many grid cells with zero values. In natural waters, though, these two nutrients are always available at least in low concentrations²², and zero values are thus misleading. Another aspect is that in our logarithmic model, zero values cannot be handled as the

Fig. 1 | Nitrogen and phosphorus in inland waters. Global gridded maps of freshwater quality classifications due to total nitrogen (TN Index) (a) and total phosphorus (TP Index) (b).

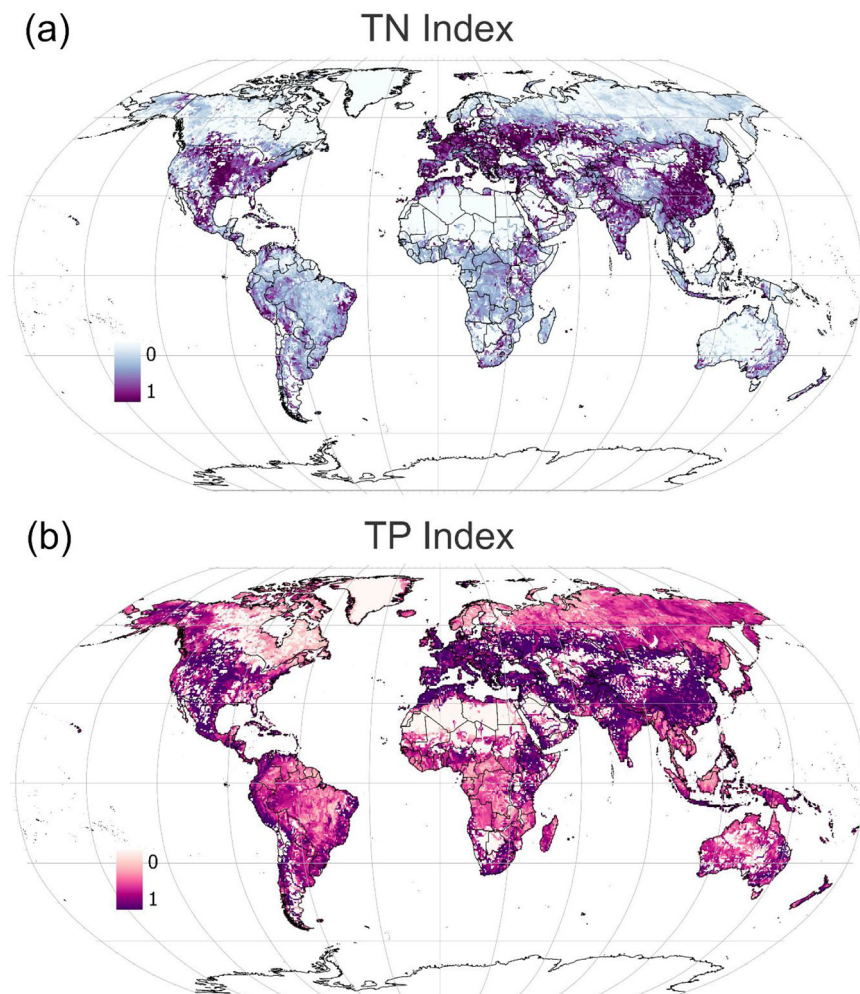
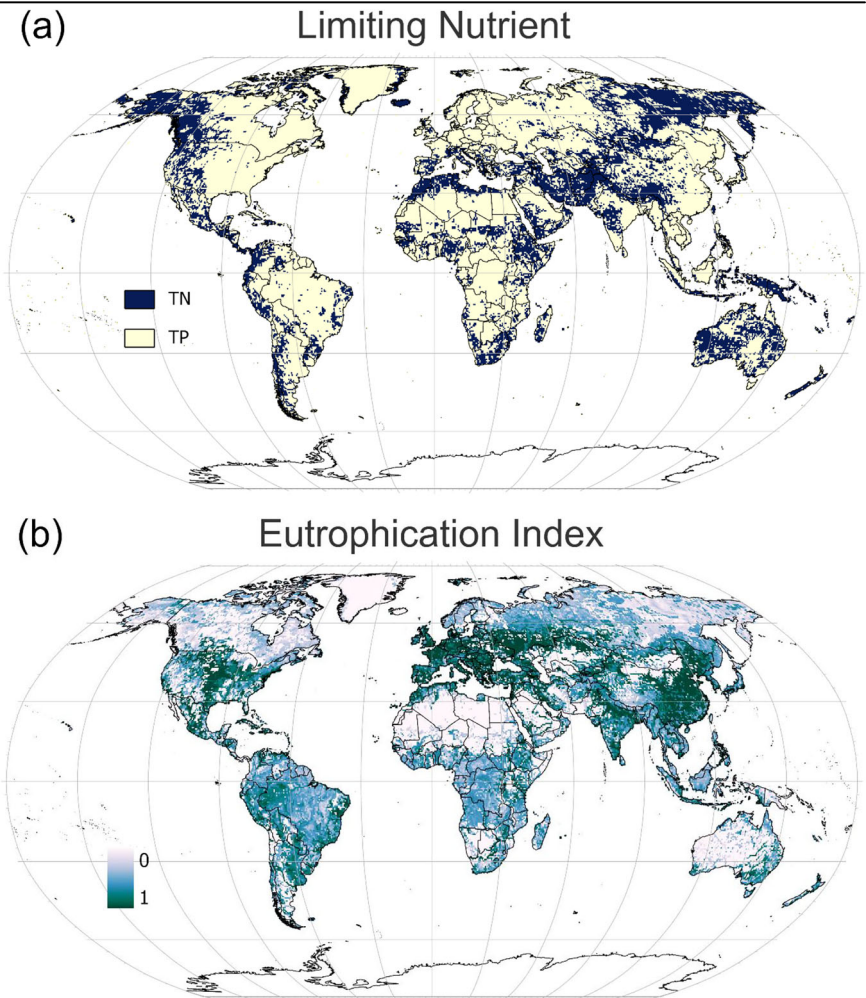


Table 1 | Trophic State Index classifications used in this study

Eutrophication class	1 Ultra-oligotrophic	2 Oligotrophic	3 Mesotrophic	4 Eutrophic	5 Hypertrophic	Unit
TP	<0.004	0.004–0.01	0.01–0.035	0.035–0.1	>0.1	mg/L
TN	<0.3	0.3–0.75	0.75–1.5	1.5–2.5	>2.5	mg/L
Attributes	Clear water, oxygen throughout the year in the hypolimnion	Hypolimnia of deeper lakes may become anoxic.	Water moderately clear; increasing probability of hypolimnetic anoxia during summer.	Blue-green algae dominate, algal scums and macrophyte problems. Notable anoxia problems.	(light limited productivity). Dense algae and macrophytes. Fish kills due to anoxia.	

These classifications have been synthesized from key international policy documents^{26–28,30}.

Fig. 2 | Global spatial distribution of nitrogen and phosphorus limitation of inland waters and eutrophication. A map for limiting nutrients for eutrophication showing areas in which TN is more important limiting factor than TP and the opposite (a), and a map for the Trophic State Index (TSI) indicating the severity of eutrophication in freshwaters (b).



logarithm of 0 is undefined. Also, calculating the N/P ratio with P being zero yields an undefined outcome. Further, the use of zero values in multiplicative risk analyses promoted widely^{32,33} yields in zero risk, which can yield to an underestimation of the risk.

TN Index = max(0, min(1, (1.3946* log(TN) + 2.5592)/5))(R² = 0.984)

TP Index = max(0, min(1, (0.9137* log(TP) + 6.105)/5))(R² = 0.997)

Using the protocol to the van Vliet et al.²⁵ data (Fig. 1a), we can see a belt of high TN index values in the northern mid-latitudes covering most of the United States and northern Mexico, and most of the corridor from North Africa and Europe (excluding the northern parts) to China and northern Indian subcontinent, plus many of the crowded and intensively farmed areas elsewhere. The lowest TN index values occurred in northern latitudes and in drylands. The areas with high TP index had similarities with the above (Fig. 1b), yet occupied larger areas. In turn, the low TP index value regions were more restricted in size, including northern latitudes and many other areas, but less than in the case of the TN index.

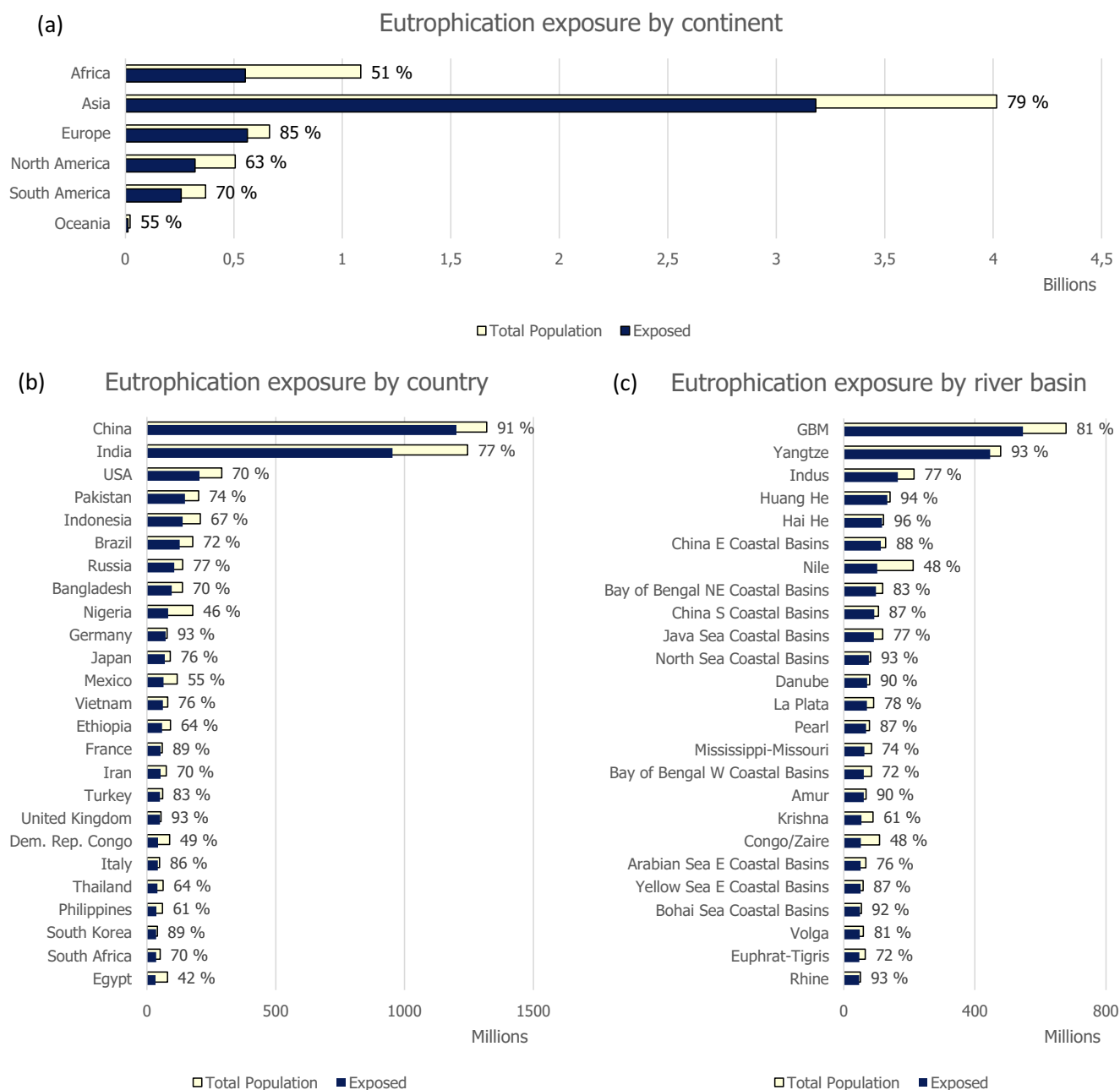


Fig. 3 | Exposure of human population to eutrophication globally. Share of people exposed (a) by continent, (b) by country and (c) by river basin. For (b), (c), only the 25 largest exposure units are shown.

The limiting nutrient. Second, we identify areas where TN being more important limiting factor for aquatic primary production than TP, and the opposite ones (Fig. 2a). As eutrophication is driven mainly by both N and P, the approach classically used in eutrophication studies is the nearly two centuries old Liebig's Law of the Minimum. It proposes that the growth of a plant is driven by the most critical factor, whatever it may be, at a given location and time. For eutrophication studies within aquatic ecosystems, the tipping point is at which the molar (atomic) N:P ratio is 16:1 (equal to 7.235 by weight as used for concentrations); above it, P is the limiting factor and below it, N limits the growth³⁴. This is commonly called the Redfield

ratio^{22,35,36}. Accordingly, we use the TP raster above the tipping point and the TN raster below it. Despite the recent evolution of more elaborate co-limitation schemes for N and P in ecosystems, Liebig's law has still been recommended to be used as a useful approximation of co-limitation⁵.

TN appeared more important in many mountainous and dry areas, whereas TP often limited eutrophication more than TN in lowlands, although much local variation existed. 69% of the grid cells of Fig. 2a exhibit TP limitation, whereas 31% have TN limitation. The observation on TN and TP limitation following such geographic patterns was an unanticipated finding, not addressed originally in the research questions. However, it may

be a remarkable finding warranting further research as similar features have been detected in terrestrial ecosystems³⁷, but not thus far in inland waters.

Co-limitation of eutrophication by nitrogen and phosphorus. Finally, the eutrophication class map (Fig. 2b) can be produced from the maps of Fig. 1a, b by using the TN index map (Fig. 1a) pixel values in areas with TN being more important limiting factor than TP, and TP index map (Fig. 1b) TP limited ones. Figure 2a indicates the geographic areas in which TN is more important limiting factor than TP, and vice versa.

Examples. As an example of the use of the protocol, we estimated the global exposure of humans to eutrophication by continent³⁸, by country³⁸, and by river basin³⁹ (Fig. 3). This yields an estimate of 73% of humanity being exposed to eutrophication. Asia, as the most populated continent, dominates with 60% of the world's total exposure. In turn, Europe has the world's highest population share of those exposed (85%), Africa having only 51%. China and India are by margin the countries with biggest exposure. 72% of the world's population live in TP-limited areas, whereas 28% live in TN-limited ones. We forward other eutrophication exposure analyses to further studies. They could include fish abundance and species, threatened species, biodiversity hotspots, and others.

We have thus far used this approach to include eutrophication in two multirisk studies, one analyzing China's eight key water-related risks⁴⁰, and the other one for a risk analysis of the world's coastal zone population⁴¹. Those studies include more insight into the applicability of the protocol in such assessments, indicating that the human exposure to eutrophication overwhelms the exposure to most other water-related risks in both cases.

Discussion

Within the water quality management and policy domain, the use of water quality classifications that estimate the harm caused by deteriorated water quality to humans and human uses of water has been the standard approach. For eutrophication, such an approach is the TSI^{26,27,29–31}. We developed a TSI-based method for producing water quality classification values for eutrophication and present also analogous schemes for BOD and TDS as Supplementary Information. Our water quality index datasets can readily be used to estimate the nuisance, exposure and risk of humans and human uses to those water quality problems. Such a straightforward methodology has not been available before. By employing TSI, the methodology is compatible with standardized policy approaches, and, besides, with the evolving Earth observation-based approaches, typically relying also on TSI^{19–21}.

The co-limitation concept allows consideration of other fractions of N and P besides TN and TP. For instance, the orthophosphate would be a remarkably better estimation for biologically available P than TP^{3,22}, but gridded datasets of it are not available on a global scale. Similar limitations occur to other fractions of N and P, such as the often-used lumped indicators DIN and DIP and their ratio³⁶. The inclusion of other limiting factors, such as Silica, can be included with the same logic as what we have shown here, although we would rather pay attention to looking closer at N-fixing Cyanobacteria; their co-limitation scheme would differ notably from that of other phytoplankton⁴². For the inclusion of factors such as water temperature and irradiance, we would recommend employing approaches such as the classical multiplicative use of the Michaelis-Menten kinetics, frequently used in water quality models⁴². A discussion of the reliability of our protocol is presented in the Supplementary Information. Thus far, it is preconditioned by the reliability of the TN and TP data, but we recommend developing hybrid approaches linking nutrient concentration datasets with Earth observations to allow more rigorous uncertainty analyses.

Compared with traditional approaches that rely on in situ sampling of water quality parameters to calculate the TSI, the widespread availability of Earth observation data such as remote sensing^{19–21}, together with the rapid growth of publicly accessible water quality databases^{6,23–25}, effectively overcomes many limitations of manual monitoring, including sparse sampling sites, high costs, and the difficulty of obtaining long-term continuous time series. This allows now to quantitatively assess the degree of eutrophication across different spatial and temporal scales. In this context, given that eutrophication is one of the major water-related risk factors^{1–3,41}, affecting large populations and posing substantial ecological and public health risks, incorporating eutrophication indicators into global water quality assessment frameworks, including SDG 6, is of great importance for improving water quality monitoring capacity, strengthening water risk identification systems, and enhancing access to clean water.

Data availability

The data are available on request from the authors.

Code availability

No custom code was developed in the study.

Olli Varis  & Dandan Zhao 

¹Water and Development Research Group, Aalto University, Espoo, Finland. ²Institute of Surface-Earth System Science, School of Earth System Science, Tianjin University, Tianjin, China.

✉ e-mail: olli.varis@aalto.fi; dandan.zhao0212@outlook.com; zhaodandan@tju.edu.cn

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Author contributions

O.V. has been responsible for all aspects of the work, from concept and data to writing, via design, data analysis, drafting, finalization, and accountability. D.Z. has written, edited, and reviewed the manuscript.

Competing interest

O.V. and D.Z. are the Editorial Board Member and the Associate Editor of npj Clean Water, respectively. Neither of them was involved in the journal's review of, or decisions related to, this manuscript. Besides, both authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this manuscript.

Additional information

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Correspondence and requests for materials should be addressed to Olli Varis or Dandan Zhao.

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